

Cake Cutting Algorithms

Fair Allocation of Divisible Goods

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You may find that cake cutting is not just a piece of cake.

- *Ariel D. Procaccia: Cake cutting: not just child's play. Communicationis of the ACM, 2013.*
- *Jeff Edmonds & Kirk Pruhs: Cake cutting really is not a piece of cake. TALG 2011.*



Outline

- 1 Motivation and Model
- 2 Classic Cake Cutting Algorithms
- 3 Query Complexity
- 4 Optimal Cake Cutting
- 5 Price of Fairness



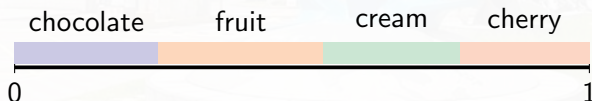
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Why Cake Cutting?

- The “cake” is a heterogeneous divisible resource: land, time, bandwidth, advertising slots, or computational resources.
- Agents may value different parts of the cake differently.
- **Goal:** design protocols that give formal fairness guarantees using limited information.



Computational Social Choice Viewpoint

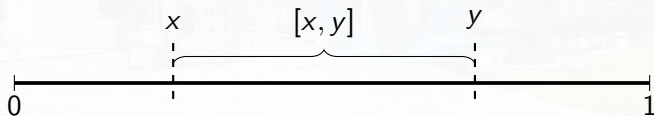
Two concerns interact

- ① **Normative:** Which fairness properties should the allocation satisfy?
 - ② **Algorithmic:** How many operations or queries are needed to find such an allocation?
- Cake cutting is a clean testbed for continuous-input fair division.
 - Cake cutting also exposes the trade-off between fairness and efficiency: enforcing fairness may reduce the maximum achievable social welfare.



Basic Setting

- Agents: $N = \{1, 2, \dots, n\}$.
- Cake: the interval $[0, 1]$.
- Agent i has a valuation function V_i over intervals.
- We write $V_i(x, y)$ for $V_i([x, y])$.



$V_i(x, y)$ is agent i 's value for this interval.

Assumptions on Valuations

For every agent i , the valuation V_i satisfies:

- **Normalization:** $V_i(0, 1) = 1$.
- **Divisibility:** for any $[x, y]$ and $0 \leq \lambda \leq 1$, there exists $z \in [x, y]$ with

$$V_i(x, z) = \lambda V_i(x, y).$$

- **Nonnegativity:** $V_i(I) \geq 0$ for every interval I .
- **Additivity:** if I and I' are disjoint, then

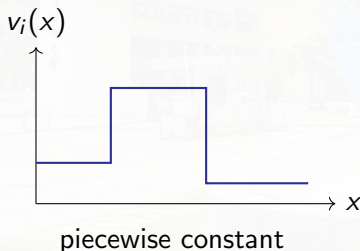
$$V_i(I \cup I') = V_i(I) + V_i(I').$$



Value Density Functions

Often V_i is induced by a density v_i :

$$V_i(X) = \int_{x \in X} v_i(x) dx, \quad \int_0^1 v_i(x) dx = 1.$$



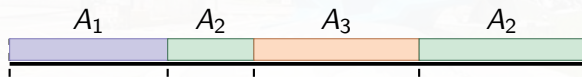
Pieces and Allocations

Definition

A **piece** of cake is a finite union of disjoint intervals.

Definition

An **allocation** is $A = (A_1, A_2, \dots, A_n)$, where A_i is agent i 's piece, the pieces are disjoint, and their union is the whole cake.



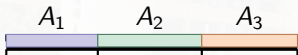
Agent 2's piece is non-contiguous here.



Contiguous vs. Non-Contiguous Allocations

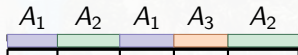
Contiguous

Each A_i is a single interval.



Non-contiguous

A piece may be a union of intervals.



Contiguity is natural in some applications, but it often makes algorithmic problems much harder.

Fairness Properties

Proportionality

Every agent receives at least its fair share:

$$V_i(A_i) \geq \frac{1}{n} \quad \forall i \in N.$$

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Equitability

All agents assign the same value to their own pieces:

$$V_{\textcolor{brown}{i}}(A_{\textcolor{brown}{i}}) = V_{\textcolor{teal}{j}}(A_{\textcolor{teal}{j}}) \quad \forall i, j \in N.$$



Envy-Freeness Implies Proportionality

Claim 1

Every envy-free allocation is proportional.



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Proof idea.

Fix agent i . Since the pieces partition the cake,

$$\sum_{j \in N} V_i(A_j) = V_i([0, 1]) = 1.$$

Hence there exists some j such that $V_i(A_j) \geq 1/n$. If the allocation is envy-free,

$$V_i(A_i) \geq V_i(A_j) \geq \frac{1}{n}.$$

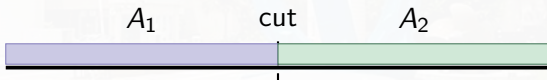


For **Two** Agents: Proportionality = Envy-Freeness

For $n = 2$, if $V_i(A_i) \geq 1/2$, then by additivity

$$V_i(A_{3-i}) = 1 - V_i(A_i) \leq 1/2 \leq V_i(A_i).$$

So agent i does not envy the other agent.



If each side is worth at least $1/2$ to its owner, no envy is possible.

For **Three or More** Agents: The Notions Split

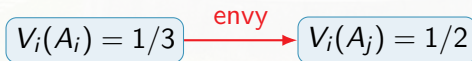
- Envy-freeness is **stronger** than proportionality.
- A proportional allocation can still generate envy.

Example pattern

For $n = 3$, agent 1 may have

$$V_1(A_1) = \frac{1}{3}, \quad V_1(A_2) = \frac{1}{2}, \quad v_1(A_3) = \frac{1}{6}.$$

For agent $j \in \{2, 3\}$, $v_j(A_1) = v_j(A_2) = v_j(A_3) = \frac{1}{3}$. Proportionality holds for agent 1, but envy-freeness fails.



Note: Equitability Does Not Imply EF or PROP

Example

	g_1	g_2
a_1	0.4	0.6
a_2	0.6	0.4

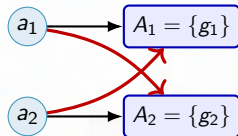
Allocate $A_1 = \{g_1\}$, $A_2 = \{g_2\}$.

- Equitable: since $v_1(A_1) = 0.4 = v_2(A_2)$.
- Not envy-free:

$$v_1(A_2) = 0.6 > 0.4 = v_1(A_1), \quad v_2(A_1) = 0.6 > 0.4 = v_2(A_2).$$

- Not proportional:

$$v_i(A_i) = 0.4 < \frac{v_i(M)}{2} = 0.5 \quad \text{for } i = 1, 2.$$



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A Nonconstructive Benchmark

Theorem 1 (Alon, 1987)

If $V_1, V_2, \dots, V_n$ are induced by continuous density functions, the cake **can be cut** in $n^2 - n$ places and partitioned into n pieces $A_1, A_2, \dots, A_n$ such that

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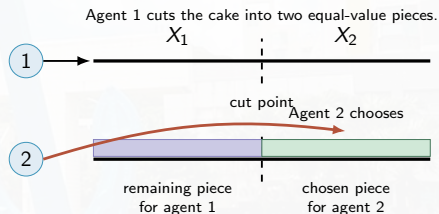
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- A very strong existence benchmark: every agent values every piece equally (equitability & envyfreeness & proportionality).
- We focus on **constructive** algorithms: how can we actually find fair allocations?
 - Achievable but it requires $n^2 - n$ **continuous** operations [Brams et al. 2006] to obtain $V_i(A_j) = \frac{1}{n}$ we need to choose cut points with arbitrary real precision (e.g., moving-knife procedures, which lie outside the discrete queue model considered later).



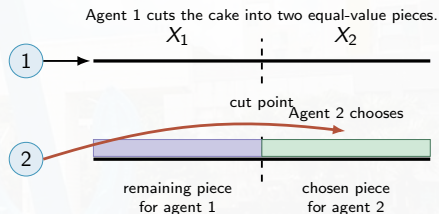
Cut and Choose: $n = 2$

- 1 Agent 1 cuts the cake into two pieces it values equally.
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- 3 Agent 1 gets the remaining piece.



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Proportional, and for two agents therefore envy-free.

Exercise: Must the Cutter Split Equally?

Consider the standard cut-and-choose procedure for two agents.

- 1 Suppose agent 1 does not know agent 2's valuation. If agent 1 wants to maximize her guaranteed utility, how should she cut the cake?

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- 4 Give an example in which agent 1 benefits from making an unequal cut because she knows which piece agent 2 will choose.

Dubins–Spanier (1961): Proportionality for Any n

- 1 Each remaining agent marks a point making the left prefix worth $1/n$ of the original cake.
- 2 The leftmost mark wins that prefix and exits.
- 3 Repeat with the remaining cake and agents.

The leftmost mark determines the prefix to be allocated.

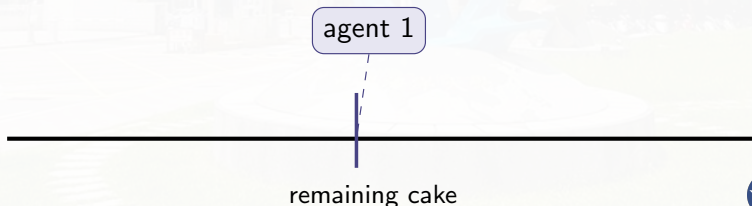
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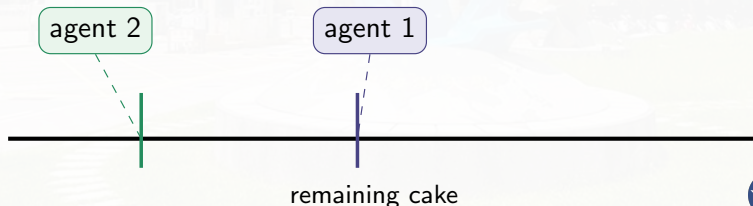
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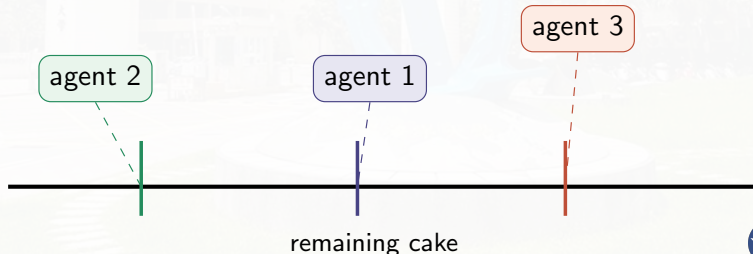
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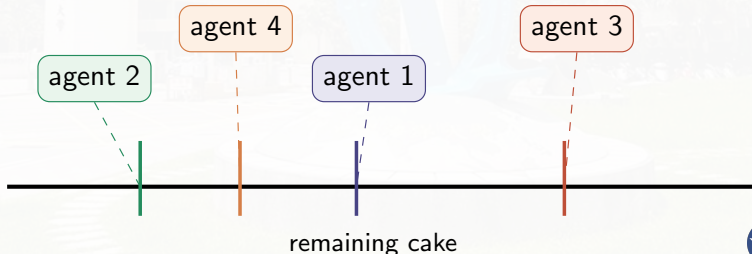
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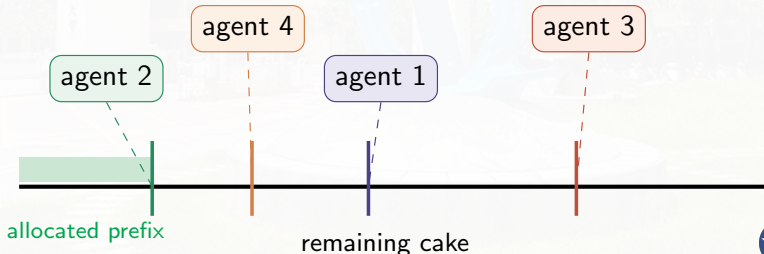
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Why Dubins–Spanier Is Proportional?

Suppose agent i^* has the leftmost mark and receives $[0, x_{i^*}]$.

- By construction, $V_{i^*}(0, x_{i^*}) = 1/n$.
- For any other remaining agent j , leftmost means $x_{i^*} \leq x_j$, so

$$V_j(0, x_{i^*}) \leq V_j(0, x_j) = 1/n.$$

- Thus no remaining agent loses more than $1/n$ value when this piece is removed.

Inductively, everyone gets value at least $1/n$ of the original cake.



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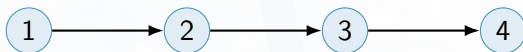
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- For the last agent j , $V_j(A_j) \geq 1 - (n-1)/n = 1/n$.



Dubins–Spanier Is Not Envy-Free



agents exiting later may receive pieces
that earlier agents prefer

- Earlier exiting agents secure value $1/n$.
- An agent would never envy earlier agents, but they may still envy a later agent's piece.



Divide the cake into three regions:

$$I_1 = \left[0, \frac{1}{4}\right], \quad I_2 = \left[\frac{1}{4}, \frac{1}{2}\right], \quad I_3 = \left[\frac{1}{2}, 1\right].$$

	I_1	I_2	I_3
1	$\frac{1}{3}$	$\frac{1}{6}$	$\frac{1}{2}$
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- Round 1: agent 1 has the leftmost $\frac{1}{3}$ -mark, so $A_1 = I_1$.

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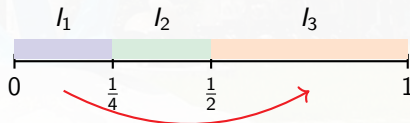


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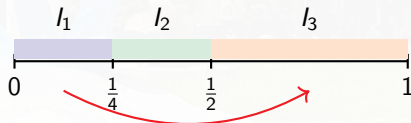
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agent 1 envies agent 3

$$V_1(A_1) = \frac{1}{3} < \frac{1}{2} = V_1(A_3).$$



Even–Paz (1984): Divide-and-Conquer

For simplicity assume n is a power of 2.

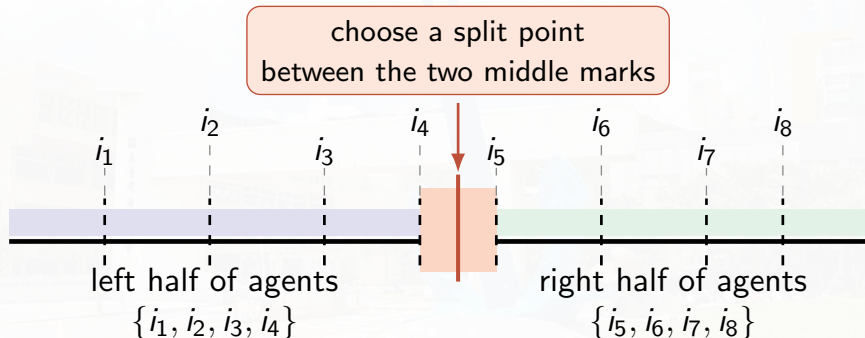
- 1 Given agents S and interval $[y, z]$, ask each $i \in S$ to mark x_i with

$$V_i(y, x_i) = \frac{1}{2} V_i(y, z).$$

- 2 Sort the marks from left to right.
- 3 Recurse on the left half of agents with the left subcake, and on the right half with the right subcake.

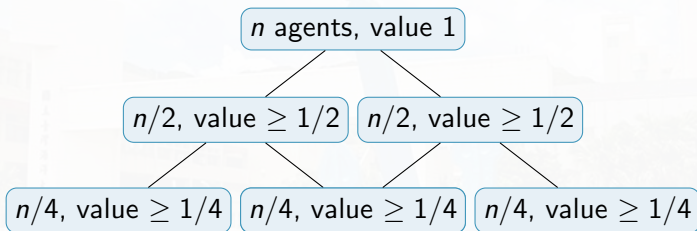


Even-Paz: One Recursive Split



The key invariant: each group of agents receives a subcake that each of them values sufficiently highly.

Even-Paz Recursion Tree



At depth k : each active agent values its current subcake at least $1/2^k$. At depth $\lg n$: each leaf agent gets value at least $1/n$.

Even–Paz: Why Proportional?

At recursive depth k :

- each active group has $n/2^k$ agents;
- every agent in that group values the group's cake at least $1/2^k$.

At depth $\lg n$, each group has one agent, and that agent receives a piece worth at least

$$\frac{1}{2^{\lg n}} = \frac{1}{n}.$$

Thus the allocation is proportional.



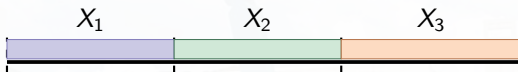
Selfridge–Conway: Envy-Freeness for $n = 3$

- First bounded envy-free algorithm for three agents.
 - Bounded number of cake-cutting operations.
- Core idea: trim the most valuable piece for agent 2 so it ties with the second-best piece.
- Allocate the main cake carefully, then allocate the trimming in a compensating order.



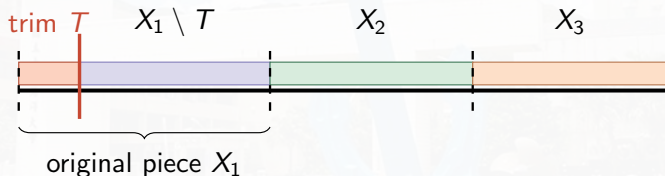
Selfridge–Conway: Initialization

- ① Agent 1 cuts the cake into three pieces it values equally: X_1, X_2, X_3 .



Selfridge–Conway: Initialization

- 1 Agent 1 cuts the cake into three pieces it values equally: X_1, X_2, X_3 .
- 2 Agent 2 trims its favorite piece (say X_1) so that it ties with its 2nd favorite.



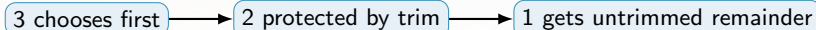
Agent 2 trims her favorite piece X_1 so that $V_2(X_1 \setminus T) = V_2(X_2)$, assuming X_2 is her second-best piece.

Now the trimmed piece and the 2nd-best piece are tied according to agent 2.



Selfridge–Conway: Allocate the Main Cake

- ③ Agent 3 chooses first from the three main pieces.
- ④ If agent 3 did not choose the trimmed piece, agent 2 gets the trimmed piece (i.e., $X_1 \setminus T$ here).
- ⑤ Otherwise agent 2 chooses from the two remaining pieces.
- ⑥ Agent 1 receives the last main piece.



Selfridge–Conway: Allocate the Trimming

Let a be the agent among $\{2, 3\}$ who received the trimmed main piece, and let a' be the other.

- ⑥ Agent a' cuts the trimming into three equal-value pieces.
- ⑦ Selection order: a , then agent 1, then a' .



Selfridge–Conway: Proof Sketch

- Main cake is envy-free:
 - agent 3 chooses first;
 - agent 2 receives a piece tied for best in its valuation;
 - agent 1 receives an untrimmed piece, which it values as one of the original thirds.
- Trimming allocation protects a and a' by the cut-and-choose logic.
- Agent 1 cannot use the trimming to envy a overall: even all the trimming would only reconstruct one original piece for agent 1.



Classic Algorithms: Comparison

Algorithm	Agents	Guarantee	Pieces
Cut and choose	2	Prop. + EF	contiguous
Dubins–Spanier	any n	Prop.	contiguous
Even–Paz	any n	Prop.	contiguous
Selfridge–Conway	3	EF	≤ 2 intervals per agent

EF = envy-free; Prop. = proportional.



Outline

- 1 Motivation and Model
- 2 Classic Cake Cutting Algorithms
- 3 Query Complexity**
- 4 Optimal Cake Cutting
- 5 Price of Fairness



Why a New Complexity Model?

Standard input-size analysis is not directly applicable:

- a valuation function over $[0, 1]$ may not have a finite discrete representation;
- algorithms often interact with agents through cuts and value reports;
- complexity is measured by the number of permitted queries.

Question

How many valuation queries are necessary and sufficient to guarantee **fairness**?



The Robertson–Webb Model

Two query types are allowed.

Evaluation query

$$\text{eval}_i(x, y) = V_i(x, y).$$

Agent i reports its **value** for interval $[x, y]$.

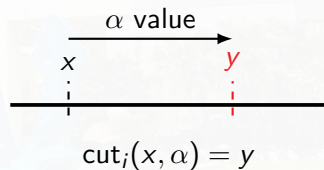
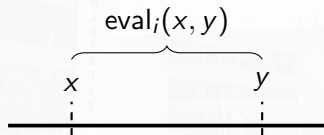
Cut query

$$\text{cut}_i(x, \alpha) = y,$$

where y is the **leftmost point** such that $V_i(x, y) = \alpha$.

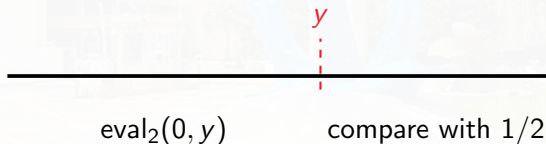


Query Types



Simulating Cut and Choose

- Ask agent 1: $\text{cut}_1(0, 1/2) = y$.
- Ask agent 2: $\text{eval}_2(0, y)$.
- If agent 2 values $[0, y]$ at least $1/2$, give it $[0, y]$; otherwise give it $[y, 1]$.



Query Complexity: Dubins–Spanier

- At a stage with k remaining agents, ask each remaining agent one cut query.

$$k = n, n - 1, \dots, 2.$$



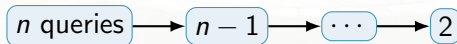
Query Complexity: Dubins–Spanier

- At a stage with k remaining agents, ask each remaining agent one cut query.

$$k = n, n - 1, \dots, 2.$$

- Thus the number of queries is

$$\sum_{k=2}^n k = \Theta(n^2).$$



Query Complexity: Even–Paz

At each recursive level, every active agent makes one cut query.

- Level 0: n queries.
- Level 1: $2 \cdot (n/2) = n$ queries.
- Level k : $2^k \cdot (n/2^k) = n$ queries.

There are $\lg n$ levels, so the total is

$$n \lg n = \Theta(n \log n).$$



Optimality of Even–Paz

Theorem 2 (Edmonds–Pruhs, SODA 2006)

Any proportional cake-cutting algorithm requires

$$\Omega(n \log n)$$

queries in the Robertson–Webb model.

Consequence

Even–Paz is **asymptotically optimal** for proportional cake cutting in this query model.



Lower Bound Strategy: Thin-Rich Pieces

We separate the problem of finding a proportional allocation into problems that must be solved **for each agent individually**.

Fix an agent i .

A piece X is:

- **thin** if $\ell(X) \leq 2/n$; ($\ell(X)$: length of the cake piece X)
- **rich** if $V_i(X) \geq 1/n$;
- **thin-rich** if both conditions hold.

Lemma 1

If **finding a thin-rich piece needs $T(n)$ queries**, then proportional cake cutting needs $\Omega(nT(n))$ queries.

Sketch of the proof of Lemma 1

- Each query only acquires information about the valuation function of a single agent, hence the interaction with one agent cannot help us find a thin-rich piece with respect to another agent.
- In any **proportional** allocation, every allocated piece **must be rich for its owner**.
- At most $n/2$ allocated pieces can have length more than $2/n$.
- Therefore, **in any proportional allocation, at least $n/2$ agents receive thin-rich pieces.**



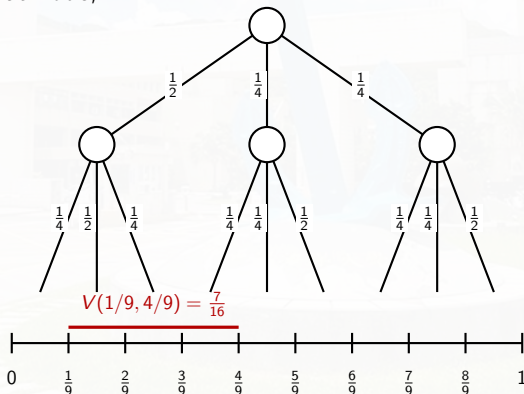
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- At most $n/2$ allocated pieces can have length more than $2/n$.
- Therefore, **in any proportional allocation, at least $n/2$ agents receive thin-rich pieces.** $\Rightarrow$ one needs to solve the thin-rich problem w.r.t. $\geq n/2$ agents.



Valuation Function: a Value Tree

- The lower bound uses **adversarial** valuation functions represented by a ternary tree.
- An internal node: the union of the intervals at the leaves of the subtree rooted at that node,



From Thin-Rich Pieces to Thin-Rich Leaves

Lemma 2

Suppose the valuation function is derived from a value tree.

If a **thin-rich piece** can be found using $T(n)$ Robertson–Webb queries, then a **thin-rich leaf** of the value tree can be found using $O(T(n))$ queries.



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If a **thin-rich leaf** of the value tree cannot be found in $O(T(n))$ queries, then a **thin-rich piece** cannot be found in $T(n)$ queries.



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If a **thin-rich piece** can be found using $T(n)$ Robertson–Webb queries, then a **thin-rich leaf** of the value tree can be found using $O(T(n))$ queries.

- A leaf is an interval of length $2/n$.
- A piece X is thin-rich if

$$\ell(X) \leq \frac{2}{n} \quad \text{and} \quad V(X) \geq \frac{1}{n}.$$

- **Claim:** It is enough to prove lower bounds for finding a rich leaf.



Key Observation: Some Interval Has High Density

- Suppose an algorithm finds a thin-rich piece X .
- Write X as a union of intervals. Since X is rich, $V(X) \geq \frac{1}{n}$.



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Claim: The density of I^* is $\geq 1/2$

There exists an interval $I^* \subseteq X$ such that $V(I^*) \geq \frac{\ell(I^*)}{2}$.

- Otherwise, if every interval $I \subseteq X$ satisfied

$$V(I) < \frac{\ell(I)}{2},$$

then

$$V(X) = \sum_{I \subseteq X} V(I) < \sum_{I \subseteq X} \frac{\ell(I)}{2} = \frac{\ell(X)}{2} \leq \frac{1}{n},$$

contradicting that X is rich.



Finding a Thin-Rich Leaf

- Since X is thin, $\ell(X) \leq \frac{2}{n}$.
- Hence the interval $I^* \subseteq X$ also has length at most $2/n$. Because each leaf has length $2/n$, we know that I^* intersects at most two leaves.
 - Note: I^* is not assumed to coincide with a leaf or to be aligned with leaf boundaries.



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- The value density is constant on each leaf. Since the average density on I^* is at least $1/2$, one of the leaves it intersects, say u , has density at least $1/2$.
- Thus, $V(u) \geq \frac{\ell(u)}{2} = \frac{1}{n}$. So, u is rich, and since $\ell(u) = 2/n$, it is also thin.



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Query cost

After $T(n)$ queries, the output piece X has only $O(T(n))$ relevant intervals. Using $O(T(n))$ additional eval queries, we can evaluate the leaves intersecting X and find such a leaf u .

Value Tree: Rich Leaves Require Heavy Edges

- Each leaf represents a cake interval of length $2/n$. Since the cake has total length 1, the value tree has $\frac{1}{2/n} = \frac{n}{2}$ leaves.
- Let the height be $H = \log_3(n/2)$. If a leaf u has $q(u)$ heavy edges on its root-to-leaf path, then

$$V(u) = \left(\frac{1}{2}\right)^{q(u)} \left(\frac{1}{4}\right)^{H-q(u)} = \left(\frac{1}{4}\right)^{H-\frac{q(u)}{2}}.$$



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- To be rich, $V(u) \geq 1/n$.

A rich leaf must have a constant fraction of heavy edges on its path.
Discovering that information costs $\Omega(\log n)$ queries.



By $2H - \lg n = \Omega(\lg n)$:

$$\left(\frac{1}{4}\right)^{H - \frac{q(u)}{2}} \geq \frac{1}{n}$$

$$\Rightarrow 4^{H - \frac{q(u)}{2}} \leq n$$

$$\Rightarrow 2^{2H - q(u)} \leq n$$

$$\Rightarrow q(u) = \Omega(\lg n).$$

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- What we are interested in is how to simulate **cut** and **eval** queries by **revealing edges of the value tree**.
- A node u is **revealed** if the weights of *all* edges of every node on the path from root to u are known to the algorithm.



From Tree Queries to Robertson–Webb Queries

The proof answers each Robertson–Webb query by revealing enough of the value tree.

- An $\text{eval}(x, y)$ query reveals **paths to the leaves** containing x and y .
- A $\text{cut}(x, \alpha)$ query reveals **paths** sufficient to identify **the cut point**.
- ★ **Maintaining the invariant: After k queries, every root-to-leaf path contains at most $2k$ edges known to be heavy.**
 - Because we have shown that $\Omega(\lg n)$ edges on the path must be known to be heavy, the theorem will directly follow.



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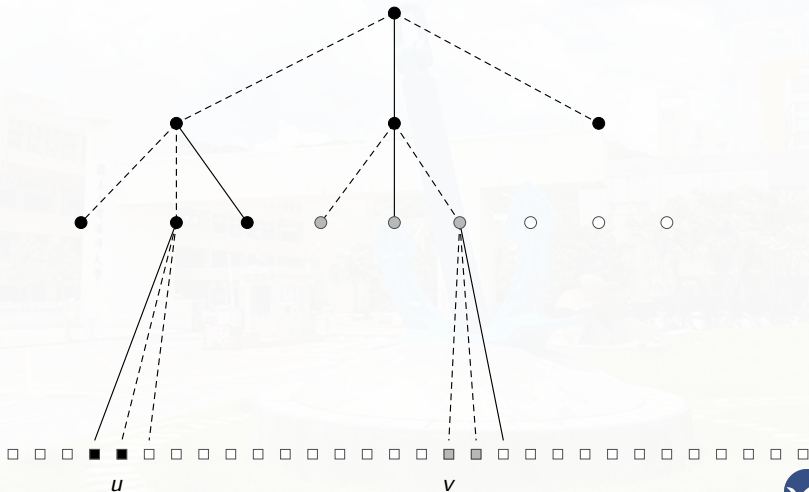
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$$c \lg n \leq q(u) \leq 2k \implies k \geq \frac{c}{2} \lg n = \Omega(\lg n).$$



Value-tree Illustration



Revealed Nodes: Useful Facts

- ★ Say V , a value function V_i of some agent i .
- If a node u is revealed, then its value $V(u)$ is known.
- If a revealed node corresponds to interval $u = [x, y]$, then $V(0, x)$, $V(x, y)$, $V(y, 1)$ are all known.
- If $z \in [x, y]$ and $[x, y]$ is a revealed leaf, then $V(0, z)$ can be computed because value density is uniform inside each leaf.
- If w lies in a revealed leaf and α is given, we can locate the leaf containing the point z satisfying $V(w, z) = \alpha$.



Why does the invariant hold? (1/2)

- Initially, the invariant holds trivially (no queries, $k = 0$, at most $2k = 0$ edges are known to be heavy)
- Assume that query $k + 1$ is an $\text{eval}(x, y)$ query.
 - Reveal the path ($r = u_0 \rightarrow u_1 \rightarrow \dots \rightarrow u_H = u$) to the leaf u containing x .
 - For every newly revealed node u_i , the edge $u_i \rightarrow u_{i+1}$ is declared **light**.
 - Other two child edges: one **light**, and one **heavy**.
- Repeat the same procedure for y .

increase of known heavy edges per query $\leq 1 + 1 = 2$.

Every root-to-leaf path now has $\leq 2k + 2 = 2(k + 1)$ known heavy edges.



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Every root-to-leaf path now has $\leq 2k + 2 = 2(k + 1)$ known heavy edges.

In a rooted tree, the intersection of any root-to-leaf path with the path to x is a prefix. Once the two paths diverge, they never meet again.



Why does the invariant hold? (2/2)

- Assume that query $k + 1$ is a $\text{cut}(w, \alpha)$. Let z be the point satisfying $V(w, z) = \alpha$.
 - Reveal the leaf containing w .
 - Find the least common ancestor u of the leaf containing w and the leaf containing z .
 - Reveal the path from u toward the leaf containing z .
- The adversary chooses (unrevealed) edge labels so that the path between u and z follows **light** edges.
- At a newly revealed node u' , let β be the remaining value needed from u' .
 - if $\beta/V(u') > 1/2 \Rightarrow$ make the left edge **heavy**;then the path must follow a middle/right **light** edge.
 - if $\beta/V(u') \leq 1/2 \Rightarrow$ make the right edge **heavy**;again the path follows a **light** edge.



Envy-Free Cake Cutting Is Harder

- Selfridge–Conway gives a bounded envy-free algorithm for $n = 3$.
- Brams–Taylor (1995) gives a *finite* envy-free algorithm for any n .
 - The algorithm is guaranteed to *terminate*.
- But for $n \geq 4$, the number of queries of Brams–Taylor is unbounded as a function of n alone.

Historical open problem

Is there a bounded envy-free algorithm for arbitrary n ?



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Is there a bounded envy-free algorithm for arbitrary n ?

Aziz–Mackenzie (STOC 2016) answered this positively: a bounded envy-free protocol exists for every n .



Bounded vs. Finite

Finite: $\forall n, \forall V, \quad Q(n, V) < \infty.$

Bounded: $\exists f : \mathbb{N} \rightarrow \mathbb{N}$ such that $\forall n, \forall V, \quad Q(n, V) \leq f(n).$



Known Lower Bounds for Envy-Freeness

Theorem (Stromquist, Electron. J. Comb. 2008)

For $n \geq 3$, no finite envy-free cake-cutting algorithm outputs contiguous allocations.

Theorem (Procaccia, AAI 2009; Theorem 3.1)

Any envy-free cake-cutting algorithm requires $\Omega(n^2)$ queries in the Robertson–Webb model.

This separates envy-freeness from proportionality:

$\Theta(n \log n)$ for proportionality vs. $\Omega(n^2)$ for envy-freeness.



Piecewise Uniform Valuations Do Not Avoid the Difficulty

Theorem (Kurokawa–Procaccia–Wang, AAAI 2013)

Let $\mathcal{A}$ be an envy-free algorithm for *piecewise uniform* valuations using at most $f(n)$ Robertson–Webb queries. Then $\mathcal{A}$ can compute an envy-free algorithm for **general** valuations using at most $f(n)$ queries.

Piecewise uniform valuations are expressive enough to capture the hardness of envy-free cake cutting.



ε -Envy-Freeness

Relax envy-freeness to

$$V_i(A_i) \geq V_i(A_j) - \varepsilon \quad \forall i, j.$$

- ① Ask each agent to cut the cake into $\lceil 1/\varepsilon \rceil$ disjoint intervals worth of value about ε .
- ② Sort all $O(n/\varepsilon)$ cut points.
- ③ Treat intervals between adjacent cut points as indivisible goods.
- ④ Allocate these small goods by round-robin selection.

This gives a simple and efficient approximate envy-free allocation.

- $O(n/\varepsilon)$ cut queries and $O(n^2/\varepsilon)$ eval queries.



Round-Robin Argument for ε -EF



- After an agent first picks, it picks before each other agent in every subsequent round.
- The only possible envy comes from goods selected earlier, each worth at most ε .

Round-Robin Argument for ε -EF



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- The only possible envy comes from goods selected earlier, each worth at most ε .
- We will talk about this algorithm more for **fair division of indivisible goods**.



Outline

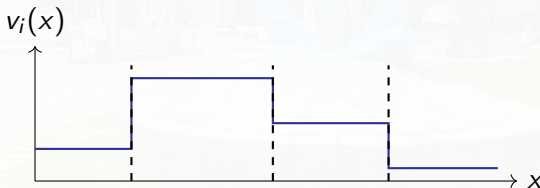
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From Query Model to Explicit Representation

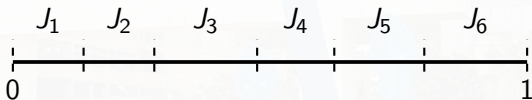
For valuations that are assumed to be explicitly represented:

- Example: piecewise constant densities.
- The input lists intervals and their density values.
- Now the computational task is to **optimize over allocations**, not to elicit valuations.
 - The notion of efficiency: **Social Welfare**.



Common Refinement of Segments

Collect all boundaries reported by all agents, plus 0 and 1. Let $\mathcal{J}$ be the set of intervals between consecutive marks.



Crucial property

For each $J \in \mathcal{J}$ and each agent i , the density v_i is constant on J .

Computing Equitable / Envy-Free Allocations

If each refined interval J is split into n equal-length subintervals and each agent gets one subinterval from every J , then every agent values every final bundle equally:

$$V_i(A_j) = \frac{1}{n} \quad \forall i, j.$$



This is constructive under explicit piecewise constant valuations.

Social Welfare

The **utilitarian social welfare** of allocation A is

$$\text{sw}(A) = \sum_{i \in N} V_i(A_i).$$

- Once welfare is introduced, fairness becomes a constraint in an optimization problem.

Optimization target

Maximize social welfare subject to proportionality, envy-freeness, or equitability.



LP for Optimal Proportional Allocation

Let f_i^J be the fraction of refined interval J allocated to agent i .

$$\begin{aligned} \max \quad & \sum_{i \in N} \sum_{J \in \mathcal{J}} f_i^J V_i(J) \\ \text{s.t.} \quad & \sum_{i \in N} f_i^J \leq 1 \quad \forall J \in \mathcal{J}, \\ & \sum_{J \in \mathcal{J}} f_i^J V_i(J) \geq \frac{1}{n} \quad \forall i \in N, \\ & f_i^J \geq 0 \quad \forall i, J. \end{aligned}$$

- A polynomial-size linear program for explicitly represented piecewise constant valuations.



LP Constraints for Other Fairness Notions

Using the same variables f_i^J , other constraints are linear.

Envy-freeness

$$\sum_J f_i^J V_i(J) \geq \sum_J f_j^J V_i(J) \quad \forall i, j.$$

Equitability

$$\sum_J f_i^J V_i(J) = \sum_J f_j^J V_j(J) \quad \forall i, j.$$

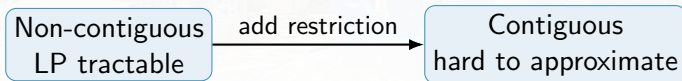
Thus LPs can compute optimal fair allocations under non-contiguous pieces.



Contiguity Makes Optimization Hard

Theorem (Bei et al., AAAI 2012)

*Given explicit piecewise constant valuations, if allocations must be both proportional and **contiguous**, then maximizing social welfare is NP-hard to obtain an $o(\sqrt{n})$ -approximation factor.*



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Price of Fairness: Definition

For a fairness property $\mathcal{F}$, define

$$\text{PoF}(\mathcal{F}) = \sup_{v_1, v_2, \dots, v_n} \frac{\max_A \text{sw}(A)}{\max_{A \in \mathcal{F}} \text{sw}(A)}.$$

- Numerator: unconstrained welfare optimum.
- Denominator: best welfare among fair allocations.
- Larger price means greater worst-case welfare loss due to fairness.



Price of Proportionality

Theorem (Caragiannis et al., WINE 2009)

The price of proportionality is

$$\Theta(\sqrt{n}).$$

Interpretation

A proportionality constraint can reduce optimal social welfare by a factor on the order of $\sqrt{n}$, and this bound is tight.



Upper Bound Idea for Proportionality

Let $\mathbf{A}^* = (A_1^*, A_2^*, \dots, A_n^*)$ be the unconstrained optimal allocation and define

$$L = \left\{ i : V_i(A_i^*) \geq \frac{1}{\sqrt{n}} \right\}, \quad S = N \setminus L.$$

- L : the set of “large” agents; S : the set of “small” agents;

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- L : the set of “large” agents; S : the set of “small” agents;
- Case 1: Many agents are “large”: $|L| \geq \sqrt{n}$. Then, we carefully redistribute pieces to protect everyone while losing at most a $\sqrt{n}$ factor.



Upper Bound Idea for Proportionality

Let $\mathbf{A}^* = (A_1^*, A_2^*, \dots, A_n^*)$ be the unconstrained optimal allocation and define

$$L = \left\{ i : V_i(A_i^*) \geq \frac{1}{\sqrt{n}} \right\}, \quad S = N \setminus L.$$

- L : the set of “large” agents; S : the set of “small” agents;
- Case 1: Many agents are “large”: $|L| \geq \sqrt{n}$. Then, we carefully redistribute pieces to protect everyone while losing at most a $\sqrt{n}$ factor.
- Case 2: Few agents are large: $|L| < \sqrt{n}$. Then, the unconstrained social welfare is already $O(\sqrt{n})$, while any proportional allocation has welfare at least 1.
 - $\text{sw}(\mathbf{A}^*) \leq |L| + |S|/\sqrt{n} < 2\sqrt{n}$, while for any proportional $\mathbf{A}$,
 $\text{sw}(\mathbf{A}) \geq \sum_{i \in N} 1/n = 1$.



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Price of Proportionality: Case 1 Setup

Case 1

Assume

$$|L| \geq \sqrt{n}.$$

Then

$$|S| = n - |L| \leq n - \sqrt{n}, \quad \sqrt{n} + |S| \leq n.$$

- Agents in L get “large” value in the optimal allocation.
- We transform $\mathbf{A}^*$ into a **proportional allocation** $\mathbf{A}$.
- Goal:

$$V_i(A_i) \geq \frac{V_i(A_i^*)}{\sqrt{n}} \quad \forall i \in N.$$



Case 1: Redistributing the Pieces

We **redistribute** each optimal piece A_i^* separately.

Pieces originally owned by *small* agents

For each $i \in S$, divide A_i^* proportionally among all agents j in S :

$$V_j(A_j \cap A_i^*) \geq \frac{V_j(A_i^*)}{|S|} \quad \forall j \in S.$$

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Pieces originally owned by *large* agents

For each $i \in L$, divide A_i^* among $\{i\} \cup S$ by treating agent i as $\sqrt{n}$ identical copies. Then

$$V_i(A_i \cap A_i^*) \geq \frac{\sqrt{n}}{\sqrt{n} + |S|} V_i(A_i^*),$$

and for every $j \in S$,

$$V_j(A_j \cap A_i^*) \geq \frac{1}{\sqrt{n} + |S|} V_j(A_i^*).$$

Case 1: Why the loss is at most $\sqrt{n}$

First, $\mathbf{A}$ is proportional.

- For $i \in L$:

$$V_i(A_i) \geq \frac{\sqrt{n}}{\sqrt{n} + |S|} V_i(A_i^*) \geq \frac{1}{\sqrt{n} + |S|} \geq \frac{1}{n}.$$

- For $i \in S$:

$$V_i(A_i) \geq \sum_{k \in L} \frac{V_i(A_k^*)}{\sqrt{n} + |S|} + \sum_{j \in S} \frac{V_i(A_j^*)}{|S|} \geq \sum_{j \in N} \frac{V_i(A_j^*)}{n} = \frac{1}{n}.$$

Case 1: Why the loss is at most $\sqrt{n}$

Moreover,

- for $i \in L$:

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- For $i \in S$, $V_i(A_i^*) < \frac{1}{\sqrt{n}}$ (by definition), and $V_i(A_i) \geq \frac{1}{n}$ (by proportionality we just built).



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$$\frac{V_i(A_i^*)}{\sqrt{n}} < \frac{1}{n} \leq V_i(A_i).$$



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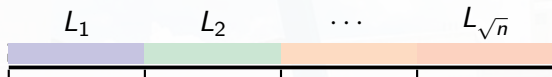
Therefore,

$$\text{sw}(A) = \sum_i V_i(A_i) \geq \frac{1}{\sqrt{n}} \sum_{i \in N} V_i(A_i^*) = \frac{\text{sw}(A^*)}{\sqrt{n}}.$$



Lower Bound Idea for Proportionality

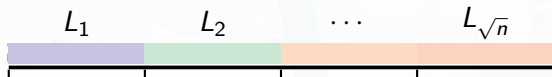
- Say that there are $\sqrt{n}$ *large* agents. Each agent $i \in L$ values **only one** distinct interval L_i of length $1/\sqrt{n}$ uniformly.



- The remaining agents (S) value the entire cake uniformly.
- Unconstrained welfare serves the large agents $\Rightarrow \text{sw}(\mathbf{A}^*) = \sqrt{n}$.
- Any proportional $\mathbf{A}$ gives an interval of length $1/n$ to each *small agent*, leaving only $(1 - (n - \sqrt{n})/n) = 1/\sqrt{n}$ to agents in L .

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$$\sum_{i \in L} V_i(A_i) + \sum_{j \in S} V_j(A_j) \leq \sqrt{n}/\sqrt{n} + 1 \leq 2.$$



Price of Envy-Freeness and Equitability

- Since envy-freeness implies proportionality, the $\Omega(\sqrt{n})$ lower bound also applies to envy-freeness.
- No nontrivial $o(n)$ upper bound on the price of envy-freeness is known.



Price of Envy-Freeness and Equitability

- Since envy-freeness implies proportionality, the $\Omega(\sqrt{n})$ lower bound also applies to envy-freeness.
- No nontrivial $o(n)$ upper bound on the price of envy-freeness is known.
- Equitability can be much more expensive.

Theorem (Caragiannis et al., WINE 2009)

The price of equitability is $\Theta(n)$.



Equitable vs. Envy-Free Welfare

Theorem (Brams et al., AAI 2012)

For piecewise constant valuations, the optimal envy-free allocation has social welfare at least that of the optimal equitable allocation.

Optimal equitable
allocation

$$sw(EF) \geq sw(EQ)$$

Optimal envy-free
allocation



Proof Tool: Linear Fisher Markets

A linear Fisher market has:

- divisible goods G ;
- utilities u_{ij} for agent i and good j ;
- budgets e_i ;
- equilibrium prices p_j
- allocations f_{ij} (giving agent i a fraction f_{ij} of good j).
- utility of agent i for an allocation: $\sum_{j \in G} f_{ij} u_{ij}$.
- For all $j \in G$, $\sum_{i \in N} f_{ij} \leq 1$.



Fisher-Market View of Cake Cutting

Start from an **equitable** welfare-maximizing cake allocation

$$\mathbf{A}^* = (A_1^*, A_2^*, \dots, A_n^*).$$

Construct a linear Fisher market

- Good j corresponds to cake piece A_j^* .
- Agent i 's utility for good j is $u_{ij} = V_i(A_j^*)$.
- Budgets are identical: $e_i = \frac{1}{n} \quad \forall i \in N$.

cake pieces A_j^* $\longleftrightarrow$ divisible Fisher goods j .



From Fisher Allocation Back to Cake

- Given a Fisher allocation $\mathbf{f} = (f_{ij})_{i \in N, j \in G}$, we want a cake allocation $\mathbf{A}$ such that

$$V_i(A_j) = \sum_{k \in G} f_{jk} u_{ik} \quad \forall i, j \in N.$$

Idea

Subdivide each A_k^* so that, within every resulting interval, each agent's value is proportional to length.

- The utilities in the Fisher market can be replicated in the cake-cutting instance.



Why the Fisher Allocation Is Envy-Free?

- Let $p = (p_1, p_2, \dots, p_n)$ be **equilibrium prices**, and define $r_i^* = \max_{j \in G} \frac{u_{ij}}{p_j}$.
- By the Fisher-market property (Vazirani 2007), agent i receives only **goods maximizing utility per price** (bang-per-buck):

$$f_{ij} > 0 \quad \implies \quad \frac{u_{ij}}{p_j} = r_i^*.$$



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- For another agent k ,

$$V_i(A_k) = \sum_{j \in G} f_{kj} u_{ij} \leq \sum_{j \in G} f_{kj} r_i^* p_j = \frac{r_i^*}{n}.$$



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- Therefore $V_i(A_i) \geq V_i(A_k)$, so $\mathbf{A}$ is envy-free.



Why Welfare Does Not Decrease

- Since $\mathbf{A}^*$ is equitable, there exists $\alpha > 0$ such that $V_i(A_i^*) = \alpha \ \forall i \in N$.

Thus $\text{sw}(\mathbf{A}^*) = n\alpha$.



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$$\sum_{i \in N} p_i = 1, \Rightarrow \sum_{i \in N} \frac{1}{p_i} \geq n^2 \text{ (Cauchy-Schwarz (Exercise)).}$$

- Therefore, $\text{sw}(\mathbf{A}) \geq \frac{\alpha}{n} \cdot n^2 = n\alpha = \text{sw}(\mathbf{A}^*)$.



Takeaways

- 1 Proportionality is algorithmically well understood in the Robertson–Webb model: $\Theta(n \log n)$ queries.
- 2 Envy-freeness is strictly harder; exact bounded algorithms are much more delicate.
- 3 Explicit piecewise constant valuations enable LP-based computation of optimal fair allocations.
- 4 Fairness can significantly reduce social welfare; the price depends strongly on the fairness notion.

